TAUT CONTINUOUS HOMOMORPHISMS PRESERVE PRE-ADMISSIBILITY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The delivered context span is the author's definition of weakly pre-adic and weakly adic rings; it carries the author's citation of [Gabber-Ramero], which is resolved from a card-local bibliography.tex holding that single entry copied verbatim from the pinned my.bib shipped with the frozen source.

Editorial note. The counted proof span cites no chapter lemma; the notions of pre-admissibility and of an ideal of definition used there sit in the chapter's surrounding discussion.

Editorial note. The preamble span points at the chapter's own preamble.tex shipped with the frozen source, used verbatim as staged.

Allowed standard prerequisites (editorial).

- Tautness of a continuous homomorphism of linearly topologized rings. Same chapter, Definition definition-taut, source lines 1299-1309. Gives that the closures of the images of the I_lambda are open and form a fundamental system of open ideals of B.
- Ideal of definition of a pre-admissible topological ring, and the implications between the adic, weakly adic, admissible and pre-admissible notions. Same chapter, the discussion following Definition definition-weakly-adic, source lines 1614-1629. Fix the notion to be established for B: an open ideal whose powers eventually lie below every open neighbourhood of 0.
- Pre-adic topological rings and adic ring homomorphisms. Same chapter, Definition definition-adic-homomorphism (for the surrounding vocabulary of pre-adic rings). Background vocabulary in which the notion of an ideal of definition is used.

Lemma 0.1. *Let $\varphi : A \to B$ be a continuous homomorphism of linearly topologized rings. If φ is taut and A is pre-admissible, then B is pre-admissible.*

Proof. Let $I \subset A$ be an ideal of definition. Let $I_\lambda \subset A$ be a fundamental system of open ideals. Then the closure J of IB is open and the closures J_λ of $I_\lambda B$ are open and form a fundamental system of open ideals of B. For every λ there is an n such that $I^n \subset I_\lambda$. Observe that J^n is contained in the closure of $I^n B$. Thus $J^n \subset J_\lambda$ and we conclude J is an ideal of definition. $\square$

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